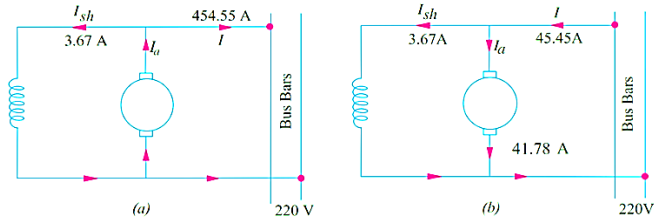
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGG.			
END SEMESTER EXAMINATION		**NOV. 2024**	
Level	**B.Tech**	**Curriculum**	**2024**
Program Name	**AUTOMOBILE ENGG.**	**Course Type**	**C/FC/PE/OE**
Course Name	**ELECTRIC VEHICLE TECHNOLOGY**	**Course Code**	**AAE 3129**
Semester	**V**	**Max Marks**	**50**
Date	**26/11/2024**	**Duration**	**3 HRS**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	Analyze the different power flow control modes in a typical parallel hybrid system, and illustrate each mode with block diagrams.	4	1	1,2,3	1	1,7	3
1B	Explain how the detailed configuration of a series hybrid vehicle affects the operational efficiency of the internal combustion engine (ICE).	3	1	7,9	1	1,7	2
1C	How would you apply the principles of torque and speed control to optimize the performance of an electric motor in an electric vehicle? Consider factors such as motor type, drive system, and power management strategies in your explanation.	3	1	2,7	1	1,7	2
2A	Draw and Discuss the Complex Hybrid and Series-Parallel Hybrid Systems	4	2	11,12	2	1,2,7	4
2B	Analyze the operation of a buck converter, impact its output voltage, efficiency, and waveform characteristics. Include wave diagrams to support your explanation	3	2,	3,7	2	1,2,7	3

2C	A belt-driven 100-kW, shunt generator running at 300 r.p.m. on 220-V bus- bars continues to run as a motor when the belt breaks, then taking 10 kW. What will be its speed? Given armature resistance = 0.025 Ohm, field resistance = 60 ohm and contact drop under each brush = 1 V, Ignore armature reaction.  	**3**	2	11,12	**2**	1,2,7	**5**
3A	Design a 3-phase inverter system for a hybrid vehicle using IGBT switches, and illustrate the corresponding waveform of the output voltage.	**4**	3	9.12	**3**	1,2,13	**5**
3B	A 4-pole series motor has 944 wave-connected armature conductors. At a certain load, the flux per pole is 34.6 mWb and the total mechanical torque developed is 209 N-m. Calculate the line current taken by the motor and the speed at which it will run with an applied voltage of 500 V. Total motor resistance is 3 ohm.	**3**	3	11,12	**3**	1,2,13	**5**
3C	How does GPS assist in real-time decision-making for transmission shift points in hybrid and electric drivetrains?	**3**	3	2.3	**3**	1,2,13	**3**
4A	Compare and contrast the power flow management strategies between the Front Hybrid Rear Electric (FHRE) and Front Electric Rear Hybrid (FERH) systems, and discuss all the 6 possible modes.	**4**	4	3,7	**4**	1,2,7	**4**
4B	How would you evaluate the performance of a lithium iron phosphate battery compared to other lithium-based battery types in a specific electric vehicle design?	**3**	4	7,9	**4**	1,2,7	**3**
4C	How do the efficiencies and performance of the following three types of fuel cells Proton Exchange	**3**	4	1,7	**4**	1,2,7	**3**

	Membrane Fuel Cells (PEMFC), Solid Oxide Fuel Cells (SOFC), and Alkaline Fuel Cells (AFC) compare?						
5A	In hybrid vehicles, multiple ECUs are often used for different subsystems. How do these ECUs communicate with each other to ensure seamless operation of the powertrain, braking, and other critical systems?	**4**	5	1,2	**5**	1,2	**3**
5B	How do mechanical couplers in a hybrid system enable regenerative braking, and what impact does this have on overall energy recovery efficiency in HEVs?	**3**	5	11,12	**5**	1,2	**3**
5C	Describe the circuit diagram and working principle of a boost converter.	**4**	3	11,12	**5**	1,2	**4**

Retain only one. Delete the others

xxxxxxx Fill in appropriately.

**** Applicable for only IET-accredited courses (modules). The IET columns may be deleted for non-IET courses.***

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***10***
2	***10***
3	***10***
4	***10***
5	***10***

BL	Total Marks
2	**06**
3	**22**
4	**12**
5	**10**